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Probability and Statistics Final Exam

Time allotted: 2:45 hrs

Exam value: 50%

Name _____

Department CO-VEA

ID No. 120

1. **TRUE** The length of time between telephone calls is continuous random variable.

1. **True** The length of time between telephone calls is exponentially distributed.
2. **False** If events X and Y are independent, then they are also mutually exclusive.

2. **False** If events X and Y are independent, then they are also mutually exclusive.

3. **True** If $P(A) = 0.3$ and $P(B) = 0.2$ and events A and B are independent, then $P(A \text{ and } B) = 0.06$.

3. **True** If $P(A) = 0.3$ and $P(B) = 0.2$ and events A and B are independent, then $P(A \cap B) = 0.06$.

4. **False** A type II error is the error that is committed if the null hypothesis is rejected when in fact it is true.

5. **False** A two-tailed hypothesis test is always more powerful than a one-tailed test.

5. ~~False~~ A two-tailed hypothesis test is always more powerful than a one-tailed test.

6. ~~False~~ The probability of committing a type I error is called level of significance (α) of the test.

1. If the mean and variance of a binomial distribution are 11 & 0.7 respectively, then the number of trials (n) and probability of success (p) for this distribution are $n =$ 30.45 & $p =$ 0.023 respectively.

2. Two events A and B are such that $P(A)=0.2$, $P(B)=0.3$, and $P(A \cup B) = 0.4$. Find the following:

$$P(A \cap B) = P(A) + P(B) - P(A \cup B)$$

$$P(A \cap B) = 0.2 + 0.3 - 0.4 = 0.1$$

a. $P(A \cap B) = 0.1$

b. $P(\bar{A} \cup \bar{B}) = 0.9$ c. $P(\bar{A} \cap \bar{B}) = \underline{0.6}$

d. $P(\bar{A}|B) = 0.66$

3. A foreign student club lists as its members 2 Canadians, 3 Japanese, 5 Italians, and 2 Germans. If a committee of 4 is selected at random,

a. The probability that all nationalities are represented **C-121**

b. The probability that all nationalities except Italian are represented

$$P(A \cap B) = P(A) \times P(B)$$

$$n \times \frac{4}{5} = 11$$

128-24

289 = 17

9 = 5/9 P = 1/9

$$p = \frac{11 - 7}{1}$$

④

~~$P(A \cap B) = P(A)P$~~

$$P(A \cup B) = \frac{P(A) + P(B) - P(A \cap B)}{0.5 - 0.4}$$

$$(A \cup B)^c = 1 - (A \cup B)$$

$$(A \cup B)^c = \begin{matrix} 1-0.1 \\ 0.6 \end{matrix} = A \cap B$$

1-2

$$\frac{P(\bar{A} \cap B)}{P(B)}$$

$$\begin{array}{r} \text{PCB} - \text{PCANB} \\ 0.3 - 0.1 \\ \underline{0.2} \\ 0.3 \end{array}$$

$$\frac{1}{(2)!}$$

$$\frac{n(E)}{\binom{12}{1}}$$

$$\binom{2}{1} \binom{3}{1} \binom{5}{1} \binom{2}{1}$$

$$\binom{2}{1} \binom{3}{1} \binom{2}{2} + \binom{2}{2} \binom{3}{1} \binom{2}{1} + \binom{2}{1} \binom{3}{2} \binom{2}{1}$$

$$\begin{array}{cccc} 2 & 3 & 2 & 1 \\ 1 & 1 & 1 & 1 \end{array}$$

1 2 3 2 12
2 3 2 2 29
1 2 2 3 12

$$\begin{array}{r} 31 + 132 + 232 \\ 6 \qquad \qquad \qquad 12 \\ \hline 495 \end{array}$$

- b. How many pages would you expect to have no misprints in a 1000-page book? (3 points)

36.78% of pages contain no misprints. So
 Out of 1000 page = $1000 \times \frac{36.78}{100}$
 = 367.8 Pages contain no
 misprints

c. What is the probability that there are at least two no misprints in a 1000-page book? (3 points)

$$\begin{aligned} \lambda &= 1 \\ P(X \geq 2) &= 1 - P(X < 2) \\ &= 1 - 0.736 \\ &= \underline{\underline{0.264}} \end{aligned}$$

$$P(x \leq 1) = P(x=0) + P(x=1)$$
$$= \frac{e^{-1} \cdot 1^0}{0!} + \frac{e^{-1} \cdot 1^1}{1!}$$
$$= 0.367 + 0.367$$
$$= 0.734$$

4. A company pays its employees an average wage of \$15.90 an hour with a standard deviation of \$1.50. If the wages are approximately normally distributed and paid to the nearest cent,

- a. What percentage of the workers receives wages between 13.5 and \$16.65 an hour inclusive? (2 points)

$$P(13.5 \leq x \leq 16.65)$$

$$\mu = 15.9$$

$\sigma = 1.5$

$$P\left(\frac{13.5 - 15.9}{1.5} \leq Z \leq \frac{16.65 - 15.9}{1.5}\right)$$

$$P(-1.6 \leq Z \leq 0.5)$$

$$P(-1.6 \leq Z \leq 0) + P(0 \leq Z \leq 0.5)$$

$$P(0 < Z < 1.6) + P(0 < Z < 0.5)$$

$$= 0.4952 + 0.1915$$

$$= 0.6367$$

$$= 63.67\%$$



- b. If 200 employees are selected from this company, approximately how many of them receive wages higher than \$14.9? (2 points)

$$P(X > 14.9) \Rightarrow P\left(Z > \frac{14.9 - 15.9}{\frac{1.5}{\sqrt{200}}}\right) = P(Z > -1)$$

$$\mu_X = \mu = 15.9$$

$$\sigma_X = \frac{\sigma}{\sqrt{n}} = \frac{1.5}{\sqrt{200}} = 0.106$$

$$P(Z > -1) = P(-1 < Z < 0) + P(Z > 0)$$

$$= P(0 < Z < 1) + P(Z > 0)$$

$$= 0.3413 + 0.5$$

$$= 0.8413 = 84.13\%$$

$$\text{out of 200} = 200 \times \frac{84.13}{100} = 168.26$$



- c. The highest 2.5% of the employee hourly wages is greater than what amount? (2 points)

$$P(X > W) = P(Z > m) = 0.025$$

$$P(Z > m) = 0.5 - P(0 < Z < m)$$

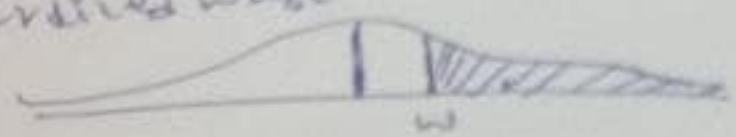
$$P(0 < Z < m) = 0.5 - 0.025$$

$$= 0.475$$

finding m on the table we get $m = 1.96$
 but this m is standardized so we have to
 back this to the normal wage

$$\frac{W - \mu}{\sigma} = m = 1.96$$

$$1.5 \times 1.96 + 15.9 = 18.84$$



the highest 2.5% of the hourly wage is greater than 18.84

- d. What is the probability that the mean hourly wage of randomly selected 16 employees will lie between \$15.15 and \$16.15? (2 points)

$$P(15.15 < X < 16.15)$$

$$\mu_X = \mu = 15.9$$

$$\sigma_X = \frac{\sigma}{\sqrt{n}} = \frac{1.5}{\sqrt{16}} = 0.375$$

$$P\left(\frac{15.15 - 15.9}{0.375} < Z < \frac{16.15 - 15.9}{0.375}\right)$$

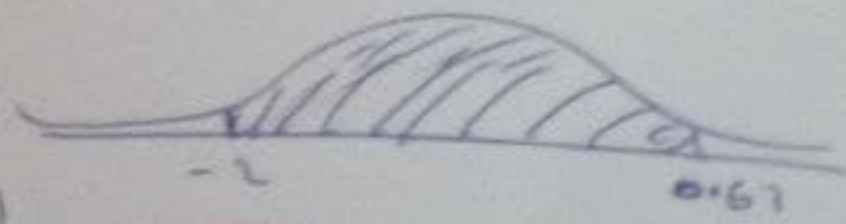
$$P(-2 < Z < 0.67)$$

$$P(-2 < Z < 0) + P(0 < Z < 0.67)$$

$$P(0 < Z < 2) + P(0 < Z < 0.67)$$

$$= 0.4772 + 0.2486$$

$$= 0.7258$$



$$\frac{0.5 - 0.05}{0.45}$$

5. A journalist claims that all adults in her city spend an average of 30 hours per month on general reading, such as newspapers, magazines, novels, and so forth. A recent sample of 25 adults from this city showed that they spend an average of 27 hours per month on general reading. The population of such times is known to be normally distributed with the population standard deviation of 7 hours. $\sigma = 7$

- a. Determine the point estimate for the mean time spent per month such reading by all adults in this city. (1 point) The point estimate of μ of the population is $\bar{x}$ of the sample mean = 27
- b. Construct the 95% confidence interval for the mean time spent per month on such reading by all adults in this city? (3 points)

$$CI = \bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$= 27 \pm 2_{0.05} \frac{7}{\sqrt{25}}$$

$$= 27 \pm 1.96 \times 1.4$$

$$= 27 \pm 2.744$$

$$= [24.256, 29.744]$$

$$\bar{x} = 27$$

$$\alpha = 0.05$$

$$\sigma = 7$$

$$n = 25$$

The mean time spent is between 24.256 and 29.744

- c. Using the 5% significance level, would you conclude that the mean time spent per month on such reading by all adults in this city is less than 30 hours? (5 points)

1, $H_0: \mu = 30$ $H_1: \mu < 30$

2, $\alpha = 0.05$

3, $z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = \frac{27 - 30}{\frac{7}{\sqrt{25}}} = \frac{-3}{\frac{7}{5}} = -2.14$

4 $z_{cal} = \frac{-3}{1.4} = -2.14$ $z_{\alpha} = z_{0.05} = 1.645$

$z_{cal} < z_{tabulated}$
 $-2.14 < 1.645$

So we reject H_0 & conclude the mean time spent on reading is less than 30

Q III: Show the necessary steps.

1. For three persons A, B, C the chance of being selected as a manager of the firm is in the ratio of 4:1:2 respectively. The respective probabilities for them to introduce a radical change in marketing strategy are 0.4, 0.8 and 0.6.

A. Find the probability that the change does take place. (2 points)

Let $P(M)$ = change in market strategy

$$P(M) = P(M|A)P(A) + P(M|B)P(B) + P(M|C)P(C)$$

$$= 0.4 \times \frac{4}{7} + 0.8 \times \frac{1}{7} + 0.6 \times \frac{2}{7}$$

$$= 0.229 + 0.114 + 0.171$$

$$P(M) = 0.514$$

$$P(A) = \frac{4}{7}$$

$$P(B) = \frac{1}{7}$$

$$P(C) = \frac{2}{7}$$

- B. If the change does take place, find the probability that it is due to the appointment of person A? (2 points)

$$P(A|M) = ?$$

$$P(A|M) = \frac{P(M|A)P(A)}{P(M)}$$

$$= \frac{0.4 \times \frac{4}{7}}{0.514}$$

$$= \frac{0.229}{0.514}$$

$$P(A|M) = 0.446$$

$$P(M) = 0.514$$

2. Let X be a continuous random variable with probability density function

$$f(x) = \begin{cases} 1+x, & -1 \leq x < 0 \\ 1-x, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Find

- A. $P(-1/2 \leq X \leq 1/2)$ (2 points)

$$P(-\frac{1}{2} \leq x \leq \frac{1}{2}) = \int_{-\frac{1}{2}}^{\frac{1}{2}} f(x) dx = \int_{-\frac{1}{2}}^0 (1+x) dx + \int_0^{\frac{1}{2}} (1-x) dx$$

$$= x + \frac{x^2}{2} \Big|_{-\frac{1}{2}}^0 + x - \frac{x^2}{2} \Big|_0^{\frac{1}{2}}$$

$$(0+0) - \left(-\frac{1}{2} + \frac{1}{4}\right) + \left(\frac{1}{2} - \frac{1}{4}\right) - (0-0)$$

$$\frac{1}{2} - \frac{1}{8} + \frac{1}{2} - \frac{1}{8} = \frac{2}{2} - \frac{2}{8} = 1 - \frac{1}{4} = \frac{3}{4}$$

B. $E(X)$

(2 points)

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

$$= \int_{-1}^0 x(1+x) dx + \int_0^1 x(1-x) dx \Rightarrow \int_{-1}^0 (x+x^2) dx + \int_0^1 (x-x^2) dx$$

$$= \left(\frac{x^2}{2} + \frac{x^3}{3} \right) \Big|_{-1}^0 + \left(\frac{x^2}{2} - \frac{x^3}{3} \right) \Big|_0^1 \Rightarrow \left(\frac{0}{2} + \frac{0}{3} \right) - \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) - \left(\frac{0}{2} - \frac{0}{3} \right)$$

$$= 0$$

C. $V(X)$

(2 points)

$$V(X) = \int_{-\infty}^{\infty} (x - \bar{x})^2 f(x) dx \Rightarrow \bar{x} = 0$$

$$= \int_{-1}^0 x^2 (1+x) dx + \int_0^1 x^2 (1-x) dx$$

$$= \left(\frac{x^3}{3} + \frac{x^4}{4} \right) \Big|_{-1}^0 + \left(\frac{x^3}{3} - \frac{x^4}{4} \right) \Big|_0^1$$

$$= 0 - \left(-\frac{1}{3} - \frac{1}{4} \right) + \left(\frac{1}{3} - \frac{1}{4} \right)$$

$$= -\frac{1}{2} + \frac{1}{3} + \frac{1}{3} - \frac{1}{4} = -\frac{1}{4}$$

3. Assume that the number of misprints per page in a certain type of book has a Poisson distribution with a mean of one misprint per one page.

$$E(X) = \lambda = 1$$

$$P(X) = \frac{e^{-\lambda} \lambda^x}{x!}$$

- a. What percentage of pages contains no misprints? (3 points)

$$P(X) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$E(X) = \lambda = 1$$

no misprint $x=0$

$$P(X=0) = \frac{e^{-1} 1^0}{0!}$$

$$= 0.3678$$

= 36.78% of pages contains no misprint